Robotics Kinematics

BRAC University

Basics of Robotics Kinematics

Kinematics comes from two Greek roots:

- **Kine** → motion
- **Matics** → related to study or analysis

So, *kinematics* literally means **the study of motion**.

In robotics, kinematics answers a very specific and practical question:

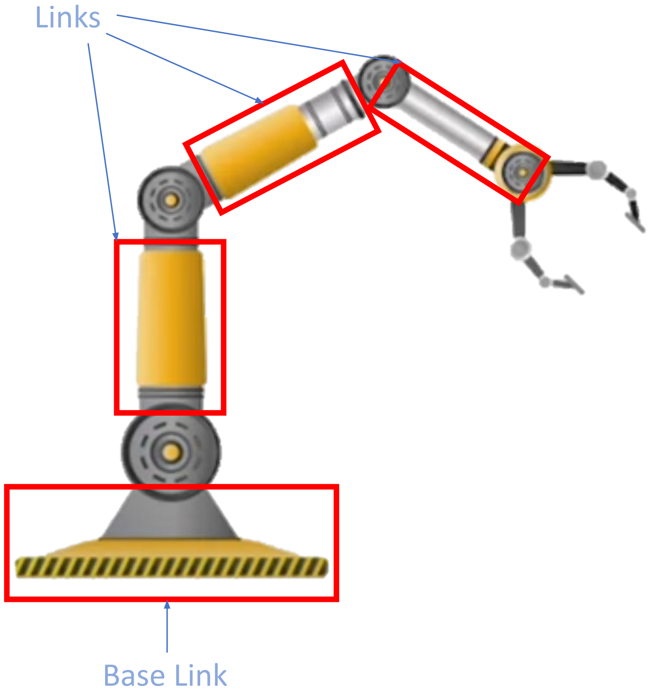
How does the motion of joints translate into the motion of the robot's end-effector?

Basics of Robotics Kinematics

Robot kinematics deals with the analytical study of the **geometry of motion** of a manipulator with respect to **a fixed reference coordinate system without regard to the force** that causes the motion.

Anatomy divided into base, links, joints and end-effector.

- **Links:** The ***rigid*** segments between the joints.
- **Joints:** The ***movable*** connections ***between links***, which can be rotational (revolute) or translational (prismatic).
- **End-Effector:** The tool at the end of the manipulator, designed to ***interact with the environment*** to achieve certain ***goals*** (e.g., grippers, welders, sensors).
- Actuators and Sensors



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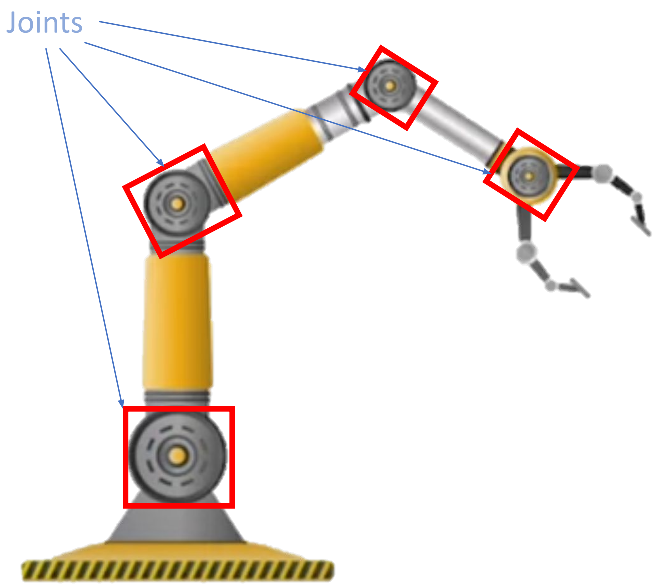

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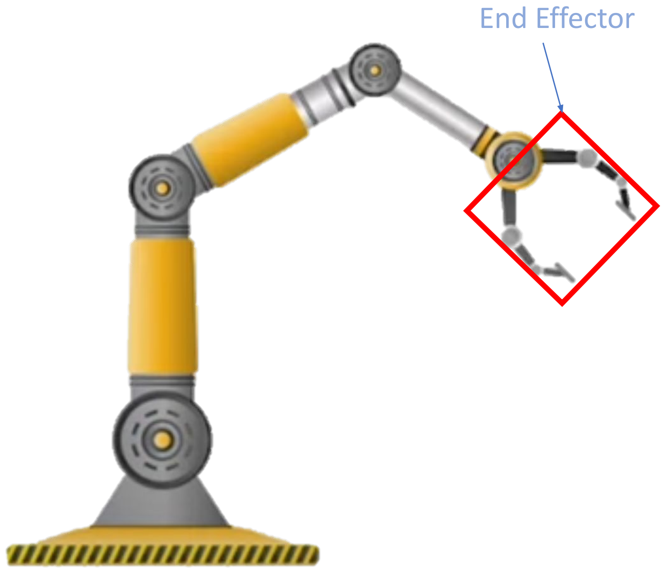


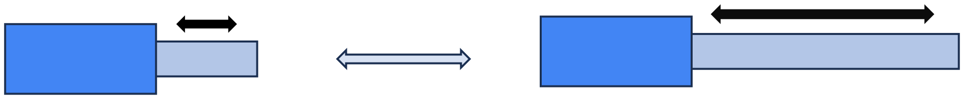
Types of Joints

Based on Mobility

- Active Joints – Can move, can change angles
- Passive Joints – Rigid. Cannot move, cannot change angles.

Based on type of movement (applicable for active joints)

- Prismatic Joints (P-joint) – Allows linear/sliding motion along a single axis (1 DoF)



- Revolute Joint (R-Joint) – Allows rotational motion around a single axis (1 DoF)

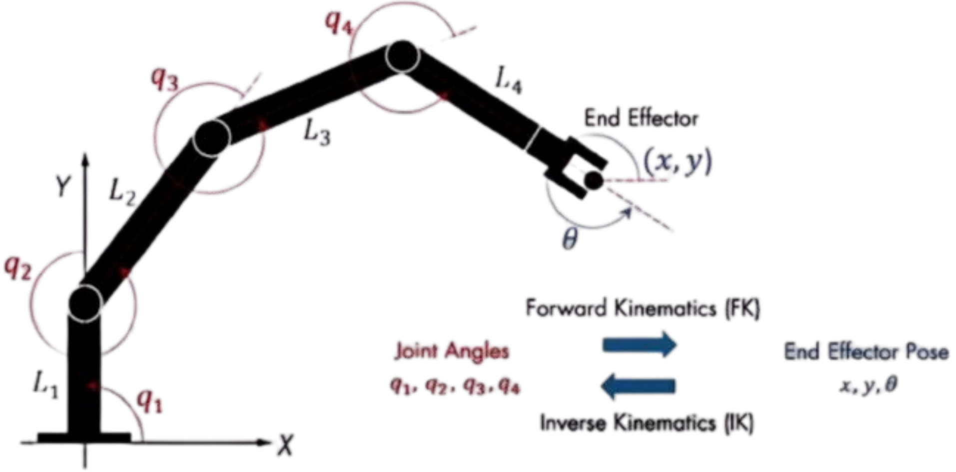
Types of Kinematics

1. **Forward/Direct Kinematics** – We want to know the final position of the end effector due to the movement of the joints.

Joint angles → x, y, θ

2. **Backward/Inverse Kinematics** – Given the final position of the end effector, we want to know the angle of the joints.

x, y, θ → Joint angles



Basics of Robotics Kinematics

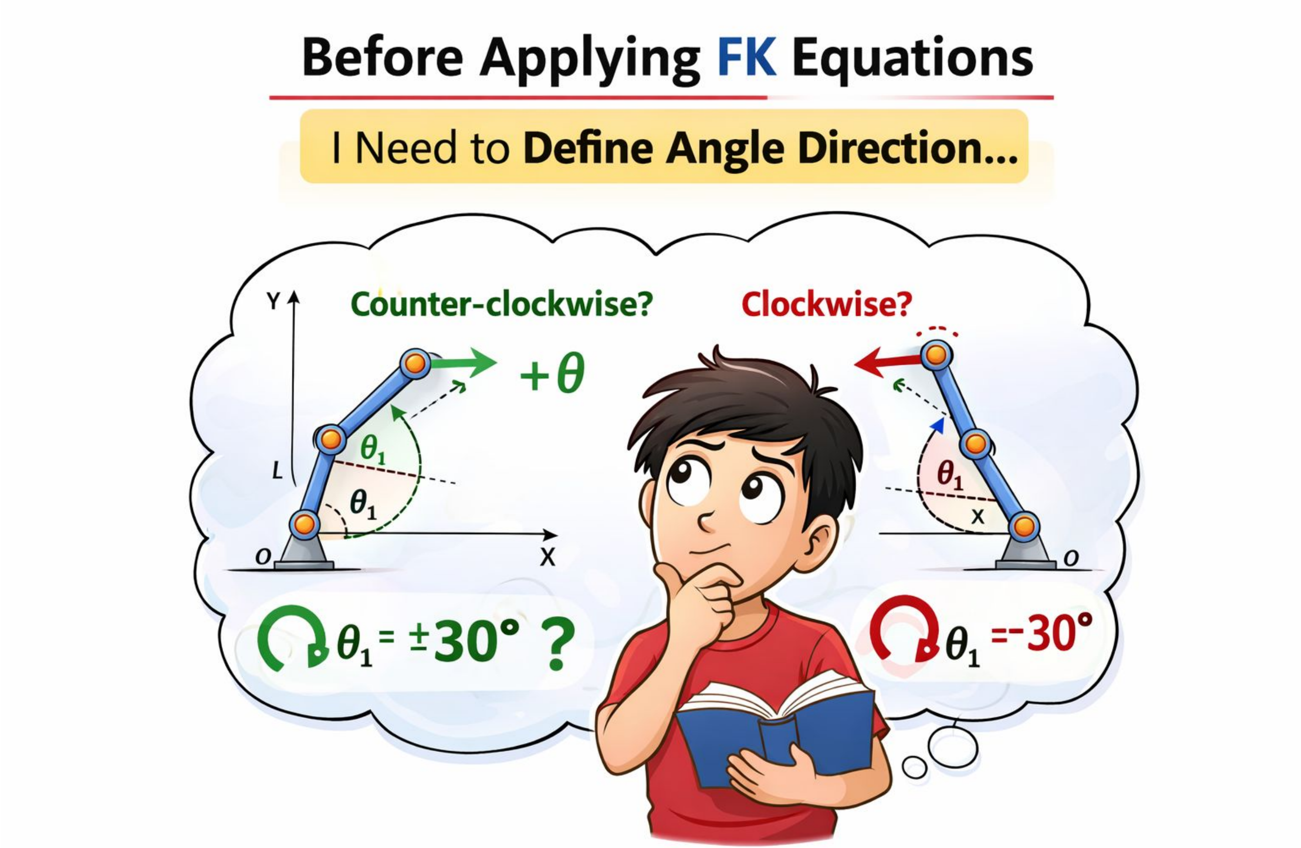
In this chapter we study how a robot arm moves in space and how joint angles relate to the position of the end-effector.

We will build everything using **pure trigonometry** before moving to the **DH-parameter** method in the next section.

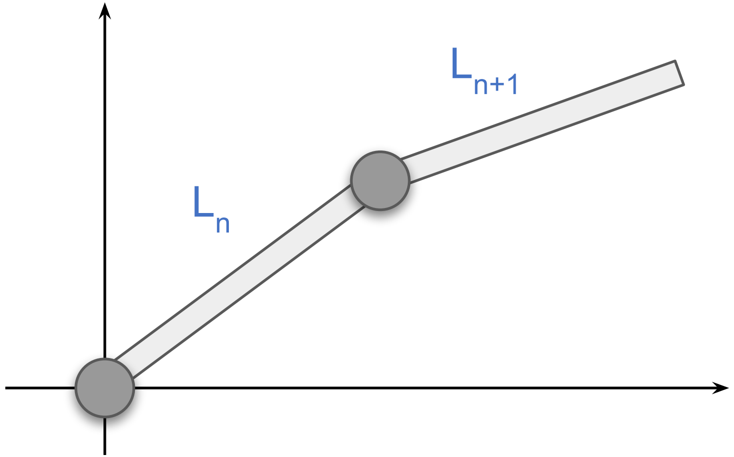
This chapter is divided into **three parts**:

1. **F**orward **K**inematics (FK) using Trigonometry.
2. **I**nverse **K**inematics (IK) using Trigonometry.
3. **F**orward **K**inematics (FK) using **D**enavit–**H**artenberg (DH) Parameters.

Preliminaries for Forward Kinematics



Angle Sign Convention

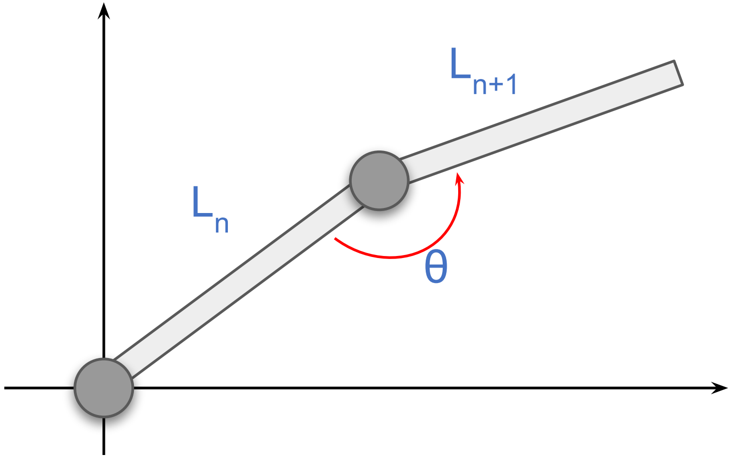


An angle is defined by **two positions**: a source (initial) position and a final (end) position.

The direction of rotation from the **initial to the end position** determines the sign of the angle:

- Counter-ClockWise **(CCW)** is positive (+)
- ClockWise **(CW)** is negative (−)

[Example] Angle Sign Convention



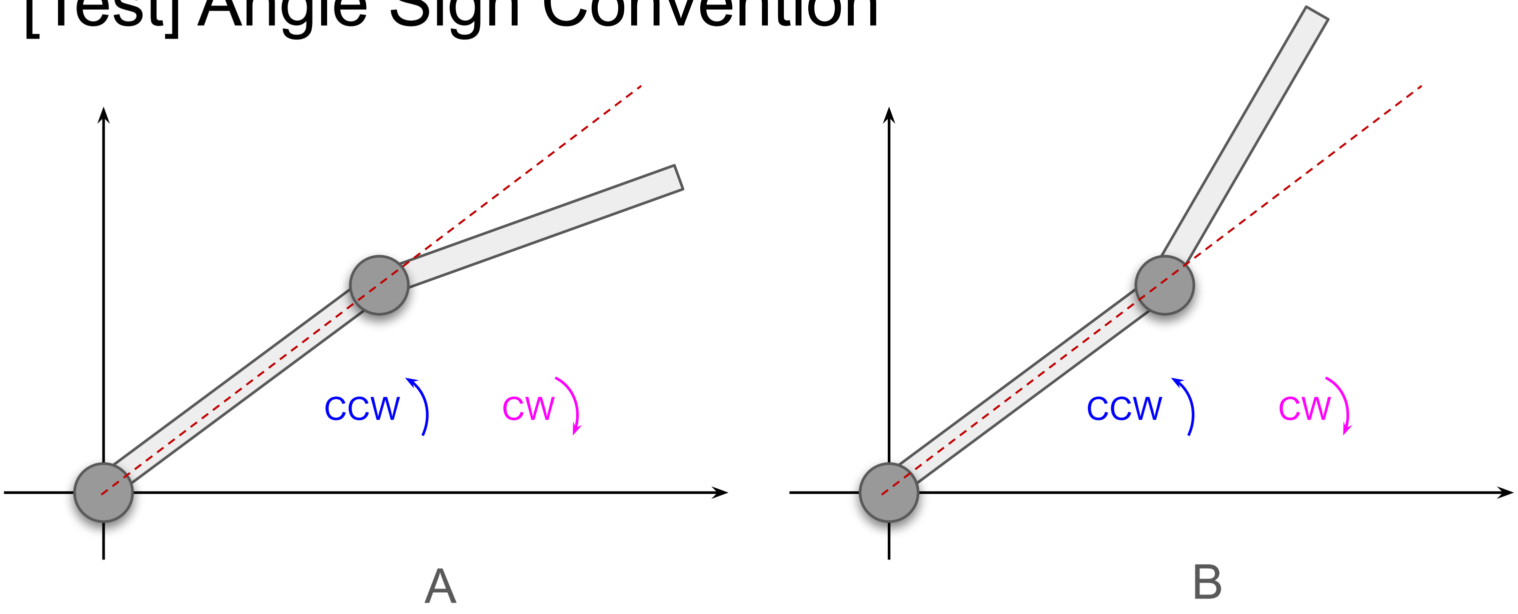
θ will be positive

Here,

L_n : Initial line or link
L_{n+1}: End line or link

So, the arrow describes the direction of the angle, the tail part of the arrow is end point.
Therefore, the direction: counter-clockwise **(CCW)**.

[Test] Angle Sign Convention



Here, the dotted line (---) is the initial line.
What will the direction of A and B figure?

[Result] Angle Sign Convention

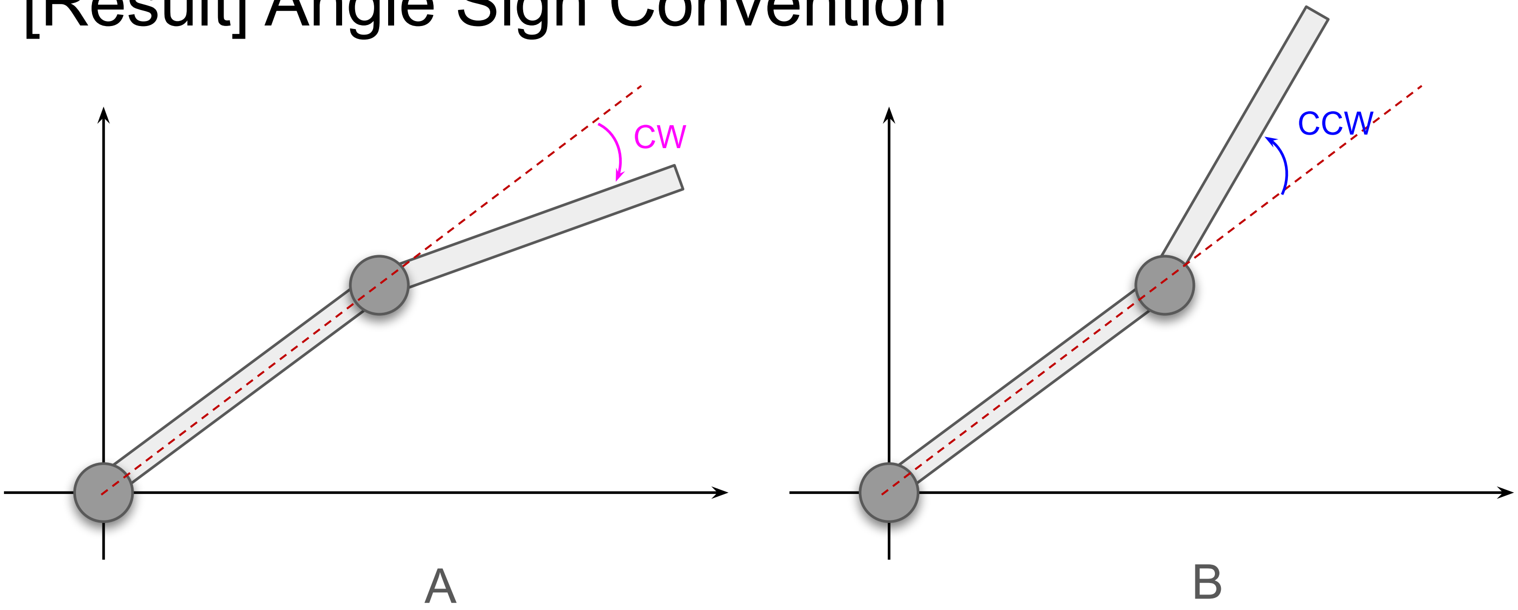


Fig. A has **negative** angle
Fig. B has **positive** angle

Forward Kinematics

Using Trigonometry

Forward Kinematics using Trigonometry

Consider a **3R Robot Manipulator** consisting,

Link lengths - L_1, L_2, L_3

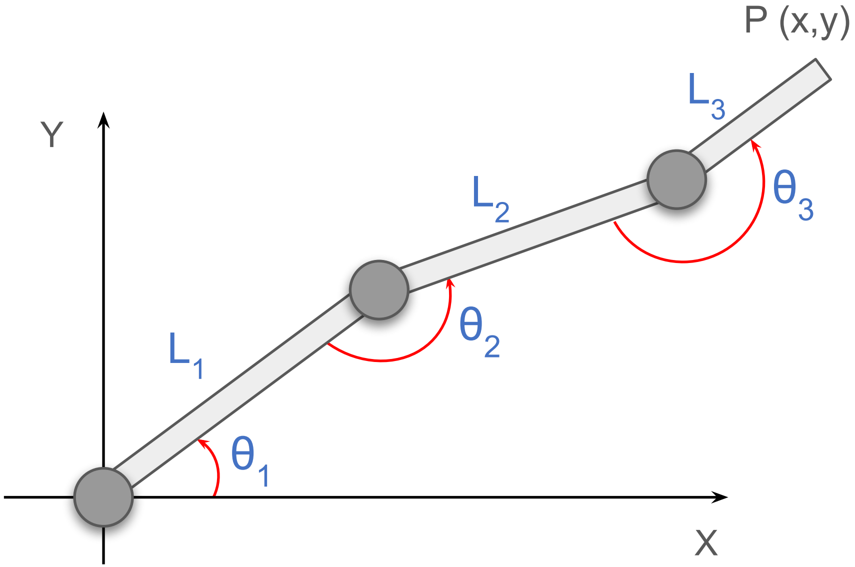
Revolute Joints - $\theta_1, \theta_2, \theta_3$

The motion of the manipulator is restricted to the two-dimensional **XY**-plane

Each joint angle is **defined relative to the immediately preceding link**, rather than with respect to the global reference frame. Specifically:

- θ_1 is measured with respect to the global X-axis
- θ_2 is measured relative to Link 1
- θ_3 is measured relative to Link 2

Our **GOAL** is to find the location of the end-effector **P(x,y)**



3R Planar Robot

https://robotics101-viscircuit.web.app/robotics/dh-model

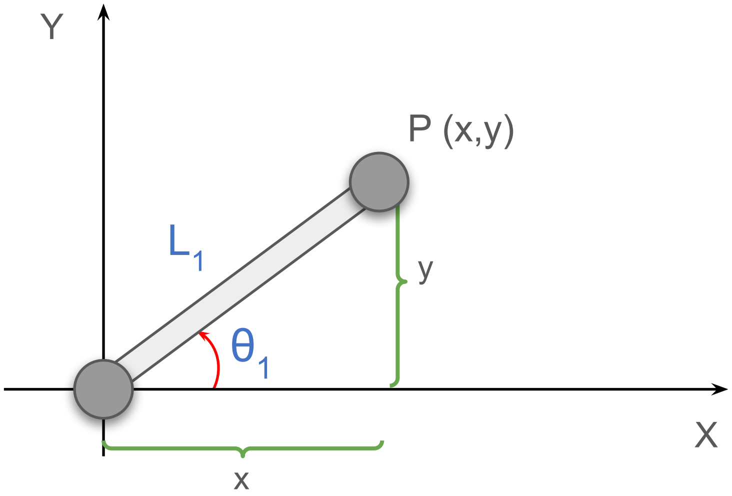
Single-Link Planar Robot

Consider a single rigid link of length $\mathbf{L_1}$ connected to the base by a revolute joint with given angle $\boldsymbol{\theta_1}$, measured from the global X-axis.

Using trigonometric formulas, the coordinates of the link endpoint are given by:

$$\mathbf{x = L_1 cos\theta_1}$$

$$\mathbf{y = L_1 sin\theta_1}$$



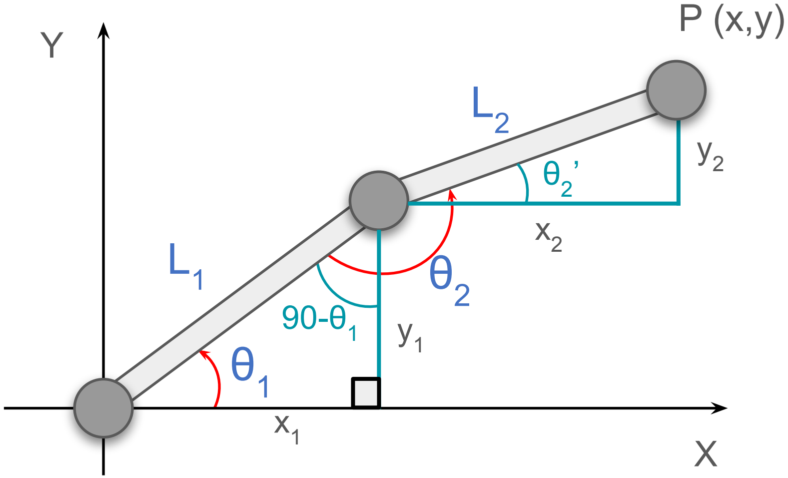
Two-Link(2R) Planar Robot

Now consider the addition of a second link of length $\mathbf{L_2}$, connected to the first link by a revolute joint $\boldsymbol{\theta_2}$

From the figure,

$$x = L_1\cos\theta_1 + L_2\cos\theta_2' \qquad [\,x_1 + x_2\,]$$

$$y = L_1\sin\theta_1 + L_2\sin\theta_2' \qquad [\,y_1 + y_2\,]$$



$$\theta_2' = \theta_2 - 90° - (90 - \theta_1)$$

$$= \theta_2 - 90° - 90 + \theta_1$$

$$= -180 + \theta_1 + \theta_2$$

Considering the value of $\boldsymbol{\theta_2'}$, we get:

$$x = L_1\cos\theta_1 + L_2\cos(-180 + \theta_1 + \theta_2)$$

$$y = L_1\sin\theta_1 + L_2\sin(-180 + \theta_1 + \theta_2)$$

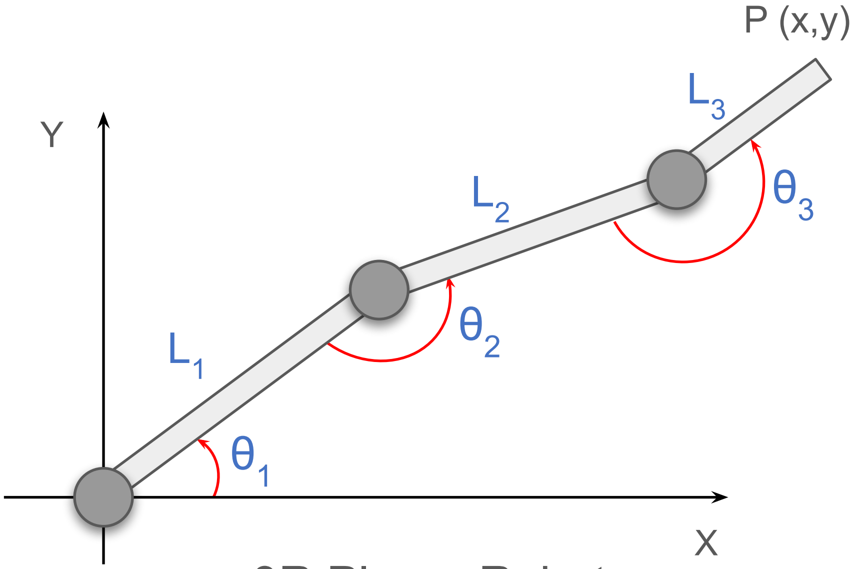
Three-Link Planar Robot (3R-2D)

Now, if θ_3 joint is also added, we get:

$$x = L_1\cos\theta_1 + L_2\cos(-180 + \theta_1 + \theta_2) + L_3\cos(-360 + \theta_1 + \theta_2 + \theta_3)$$

$$y = L_1\sin\theta_1 + L_2\sin(-180 + \theta_1 + \theta_2) + L_3\sin(-360 + \theta_1 + \theta_2 + \theta_3)$$

These equations give the **end-effector position directly from the given angles**.



3R Planar Robot

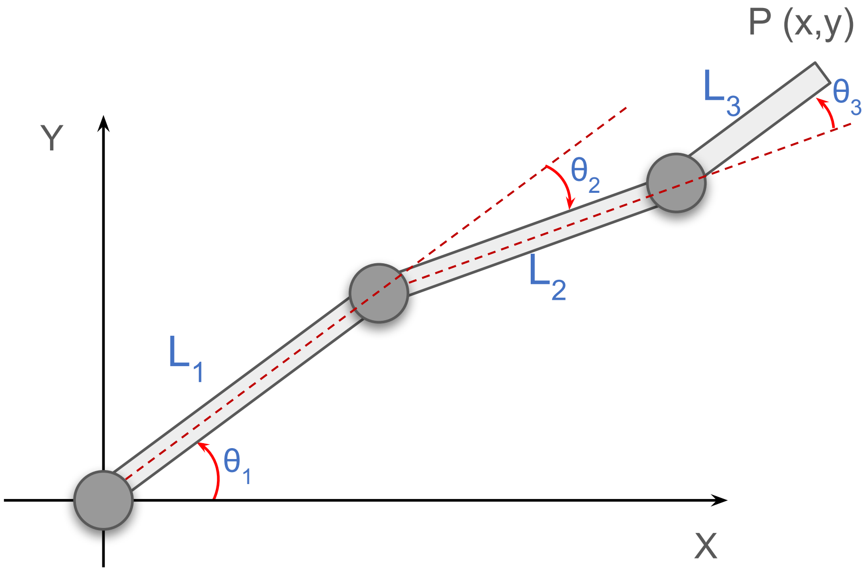
This method works, but it becomes messy as the number of joints increases. Each new joint introduces more angle bookkeeping and geometric reasoning.

Forward Kinematics using Trigonometry [Extension]

Let's say, angles are given with respect to previous link's length across [**joint angle:** in DH parameter (more details are coming in DH part)]

If we compare this with the previous figures, these angles are different.

We can modify the previous equations if we want to convert this to the earlier format.



3R Planar Robot

Forward Kinematics using Trigonometry [Extension]

Here,

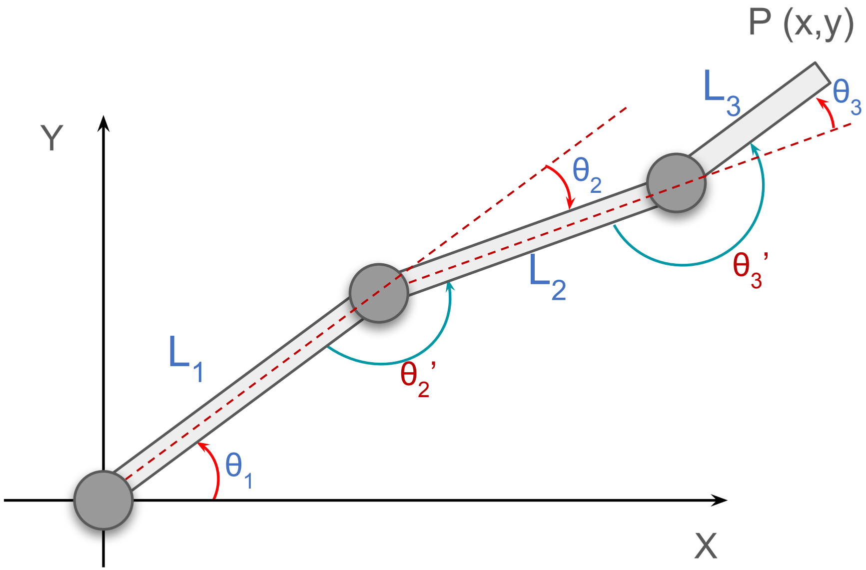
$\theta_2' = 180 + (-\theta_2)$ **[as θ_2 is CW]**

$\theta_3' = 180 + (+\theta_3)$ **[as θ_3 is CCW]**

If we plug these into the previous equation, we get:

$$x = L_1\cos\theta_1 + L_2\cos(\theta_1 - \theta_2) + L_3\cos(\theta_1 - \theta_2 + \theta_3)$$

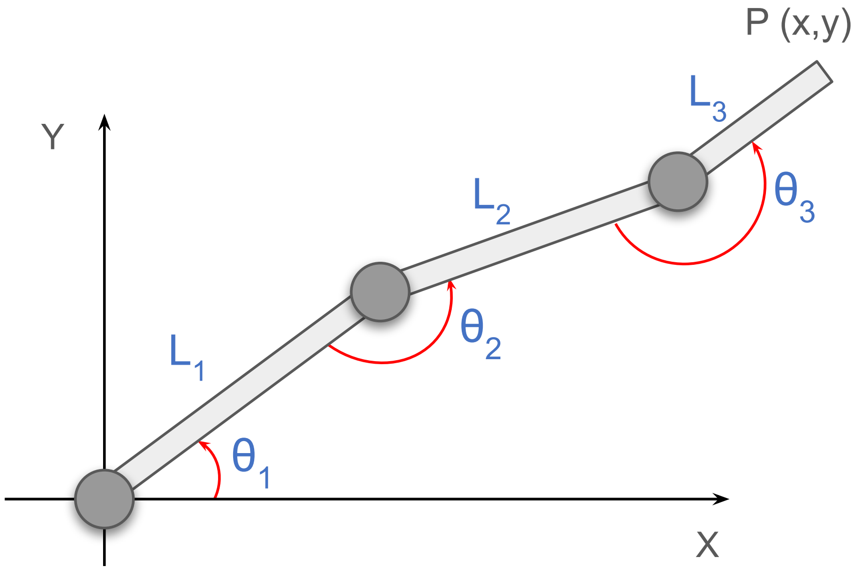
$$y = L_1\sin\theta_1 + L_2\sin(\theta_1 - \theta_2) + L_3\sin(\theta_1 - \theta_2 + \theta_3)$$



3R Planar Robot

Note: notice that **θ_1 is (+)ve, θ_2 is (-)ve, θ_3 is (+)ve. In the DH parameter we will define these angles, JOINT ANGLE**

Forward Kinematics using Trigonometry



This manipulator operates only in a single plane, does not operate in the 3 axis altogether

3R Planar Robot (3D)

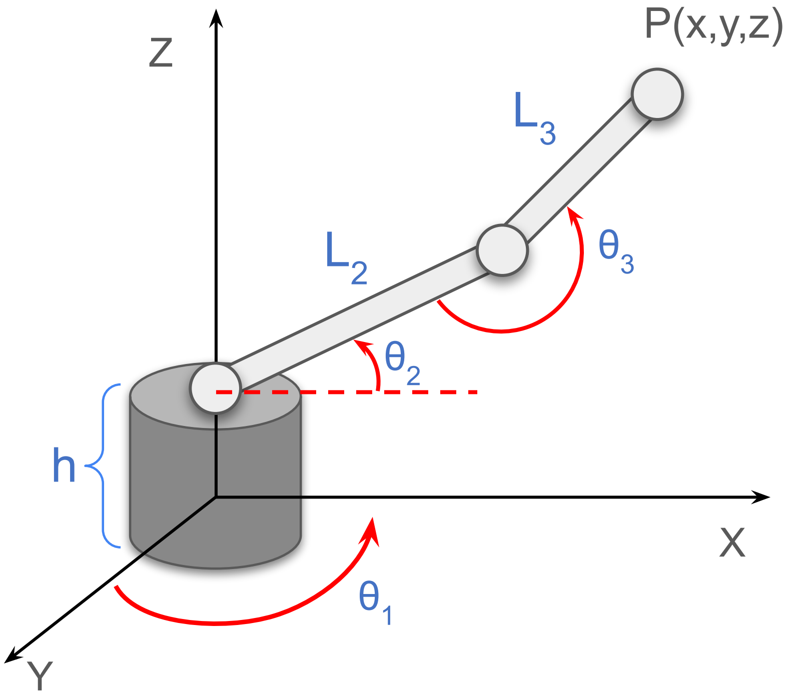
The figure illustrates a where the motion of the end-effector is not restricted to XY plane.

The base joint of this robot is configured differently, enabling rotation about a Z-axis. This modification allows the robot to access points in the *XYZ*-workspace.

GOAL - Find the location of **P(x,y,z)**

This robot can be conceptually decomposed into:

- A **base joint** that provides azimuthal motion
- A **planar 2R arm** mounted on the rotating base

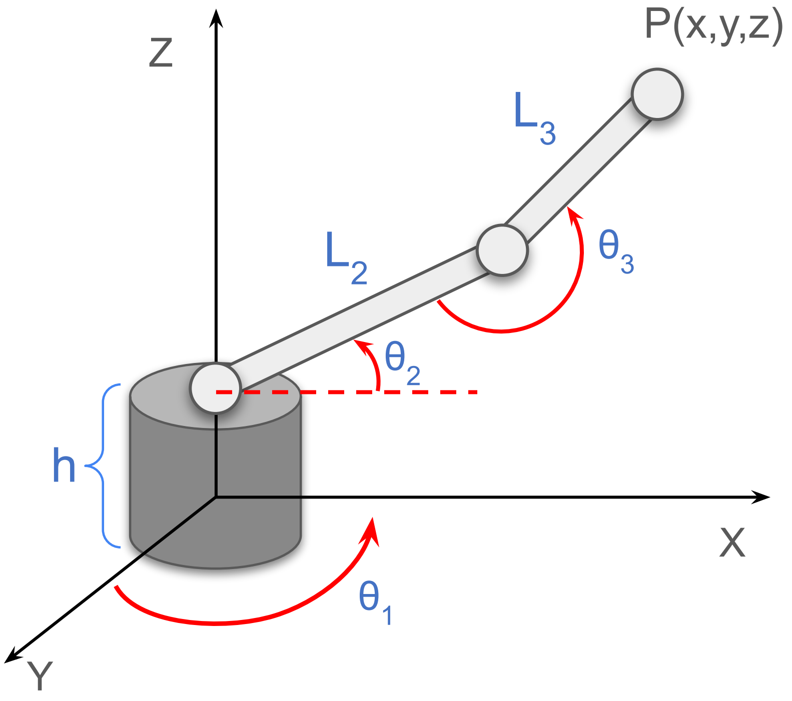


3R Planar Robot (3D)

To analyze and compute the kinematics effectively, **two orthogonal viewpoints** are required:

1. **Top View (XY-plane)**
   - Captures the effect of the base rotation
   - Used to determine the projection of motion in the horizontal plane

2. **Side View (XZ-plane)**
   - Captures the motion of Link 2 and Link 3
   - Used to compute vertical displacement and radial reach



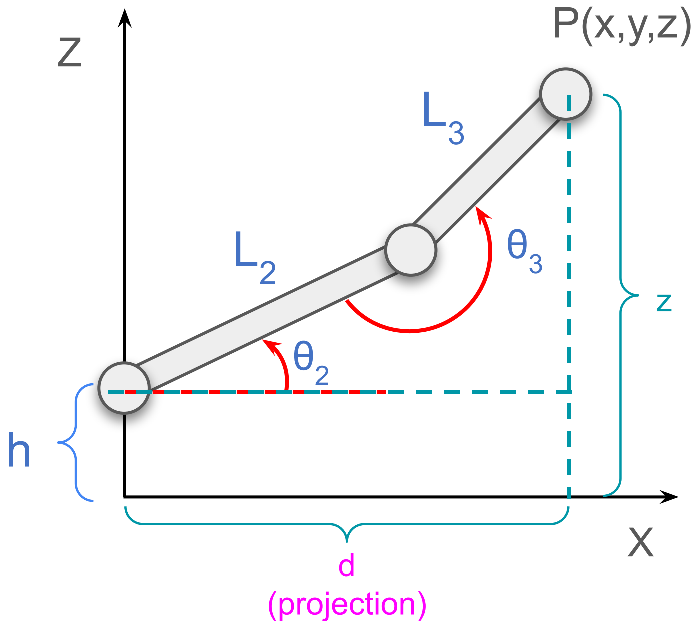
Side View

We can solve the 2R Planar part, using the earlier concepts of this lecture. The equations we get are:

$$d = L_2\cos\theta_2 + L_3\cos(-180 + \theta_2 + \theta_3)$$

$$z = L_2\sin\theta_2 + L_2\sin(-180 + \theta_2 + \theta_3) + h$$

NOTE: When analyzing the robot from the **side view**, the horizontal coordinate obtained does **not** represent the true x-coordinate of the end-effector. Instead, it represents the **projection (denoted as d)** of the arm in the horizontal plane.



Top View

From the top view, we simply see how the arm is rotated around the base and how the projected length **d** spreads into the X and Y directions.

The real coordinates,

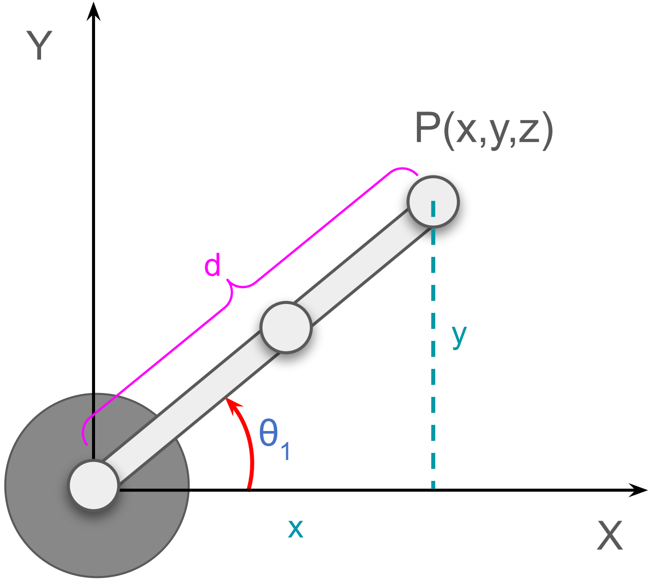
$$\mathbf{x = d\cos\theta_1}$$

$$x = (L_2\cos\theta_2 + L_3\cos(-180 + \theta_2 + \theta_3))\cos\theta_1$$

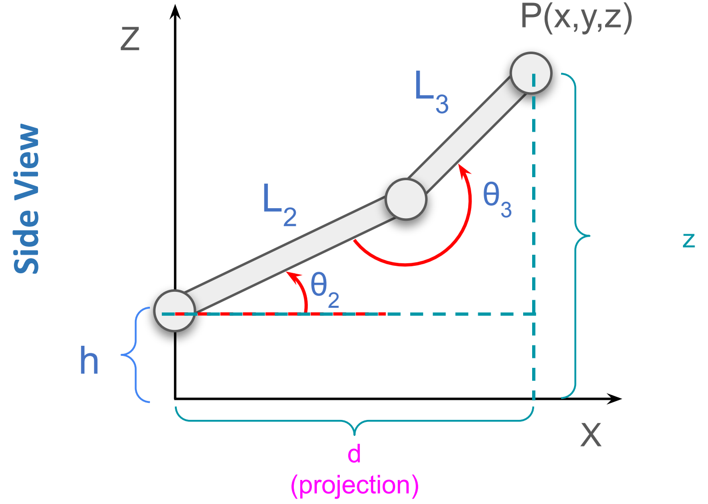
and

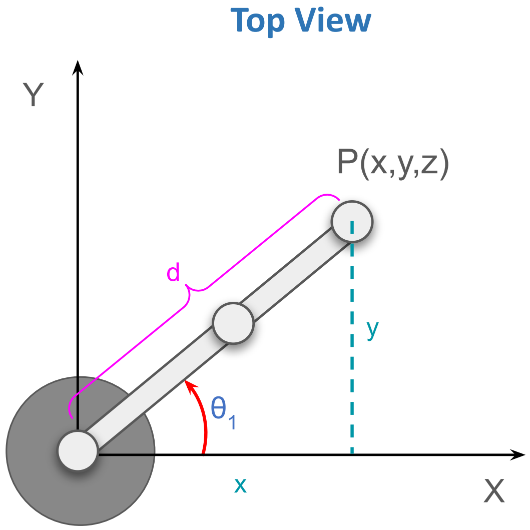
$$\mathbf{y = d\sin\theta_2}$$

$$y = (L_2\cos\theta_2 + L_3\cos(-180 + \theta_2 + \theta_3))\sin\theta_2$$



Common confusion

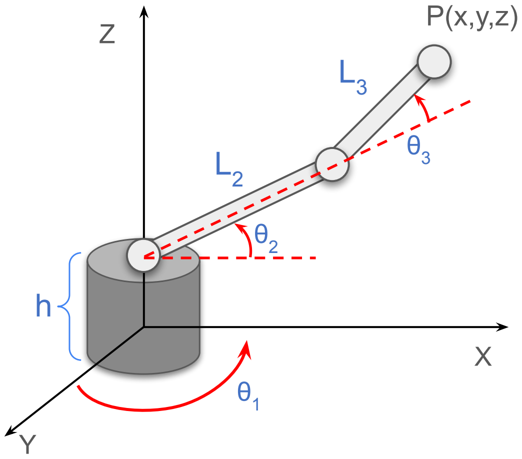


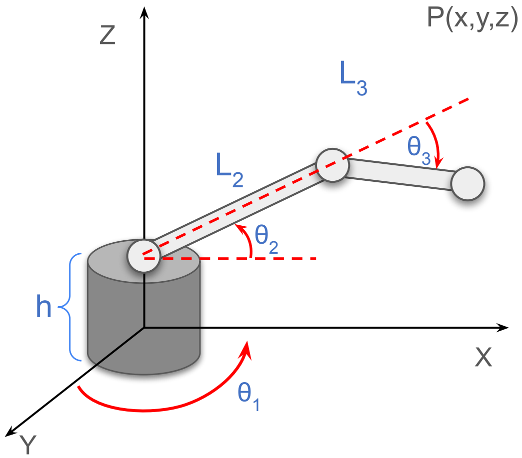


From the **side view**, d is just the **horizontal reach of the arm**, meaning how far the arm extends forward, ignoring whether it goes left or right.
This view hides the rotation of the base, so it loses direction information in the horizontal plane.
That's why we cannot get the real x-value from the side view alone. We only get the length, not where it points.

3R-3D Planar Robot [Extension]

If the angles are given with respect to previous link's length across





$$x = (L_2\cos\theta_2 + L_3\cos(\theta_2 + \theta_3))\cos\theta_1$$

$$y = (L_2\cos\theta_2 + L_3\cos(\theta_2 + \theta_3))\sin\theta_1$$

$$z = L_2\sin\theta_2 + L_3\sin(\theta_2 + \theta_3) + h$$

x = ?

y = ?

z = ?